



Week 03 Task-08

Network Scanning with Nmap

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Executive Summary

The Executive summary of this task is to perform network reconnaissance and security assessment using Nmap in an authorized home lab environment. Metasploitable 2 was used as the target system to identify open ports, running services, and service versions through different Nmap scanning techniques. The assessment helped demonstrate practical knowledge of network scanning and the importance of identifying exposed services that may introduce security risks. Additionally, the TryHackMe Nmap room was completed to strengthen understanding of advanced Nmap scanning.

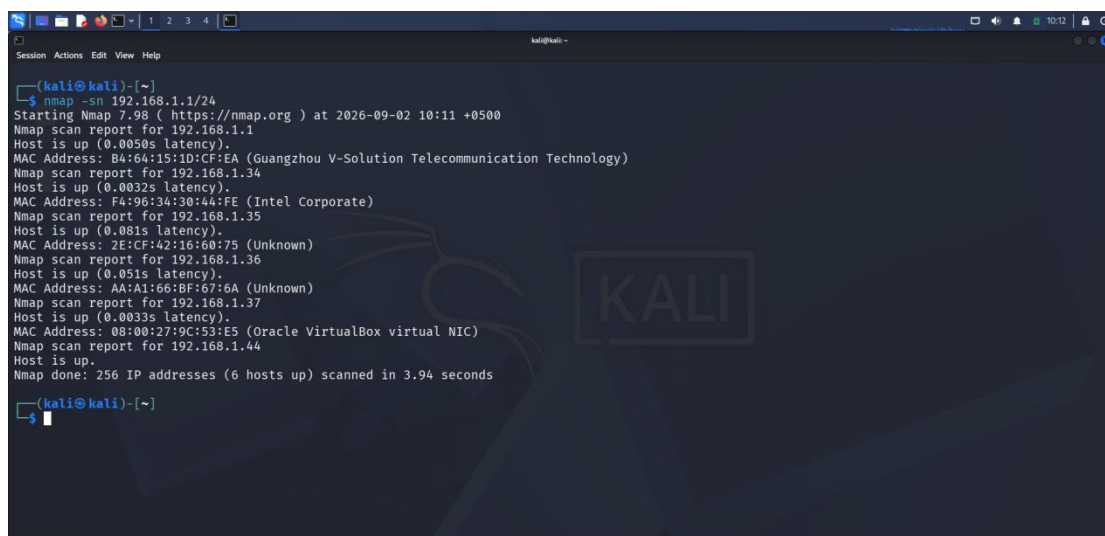
Scope

The scope of this task is to perform Nmap (**Network mapper**) scanning on an authorized home network using Metasploitable 2 as the target machine. The assessment includes host discovery, port scanning, service and version detection using -sV, and aggressive scanning techniques using A. Additionally, the TryHackMe Nmap room was completed to practice multiple scanning techniques that includes SYN, UDP, TCP, Stealth scan, Null, Fin, Xmas, and vulnerability scanning. All activities were performed in authorized lab environments only for educational and learning purposes.

Tools Used

- Kali Linux
- Nmap

Host Discovery



```
(kali@kali)-[~]
└─$ nmap -sn 192.168.1.1/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 10:11 +0500
Nmap scan report for 192.168.1.1
Host is up (0.0050s latency).
MAC Address: B4:64:15:1D:CF:EA (Guangzhou V-Solution Telecommunication Technology)
Nmap scan report for 192.168.1.34
Host is up (0.0032s latency).
MAC Address: F4:96:34:30:44:FE (Intel Corporate)
Nmap scan report for 192.168.1.35
Host is up (0.001s latency).
MAC Address: 2E:CF:42:16:60:75 (Unknown)
Nmap scan report for 192.168.1.36
Host is up (0.051s latency).
MAC Address: AA:A1:66:BF:67:6A (Unknown)
Nmap scan report for 192.168.1.37
Host is up (0.0033s latency).
MAC Address: 08:00:27:9C:53:E5 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.44
Host is up.
Nmap done: 256 IP addresses (6 hosts up) scanned in 3.94 seconds

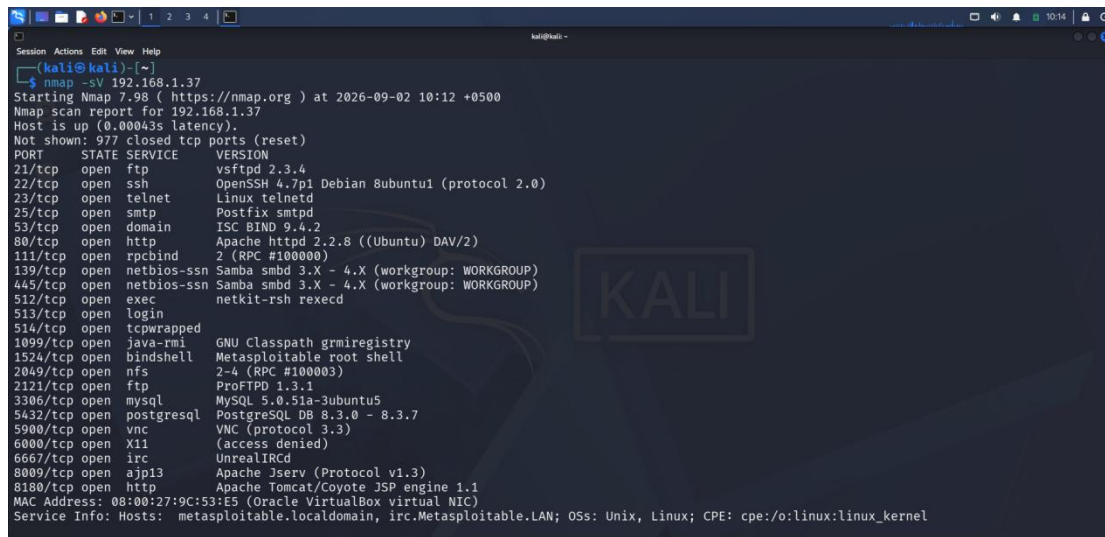
(kali@kali)-[~]
└─$
```

Figure 1.1

In the above Figure 1.1 host discovery was performed using command **nmap -sn 192.168.1.1/24**. The result obtained is explained below

S.No	HOST	Description
1	192.168.1.1	Router's IP
2	192.168.1.34	Intel Corporate
3	192.168.1.35	unkown
4	192.168.1.36	unkown
5	192.168.1.37	Metasploitable2
6	192.168.1.44	unknown

Service Version Detection



```

kali@kali:~$ nmap -sV 192.168.1.37
Starting Nmap 7.93 ( https://nmap.org ) at 2026-09-02 10:12 +0500
Nmap scan report for 192.168.1.37
Host is up (0.00043s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          netkit-rsh rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:9C:53:E5 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

```

Figure 1.2

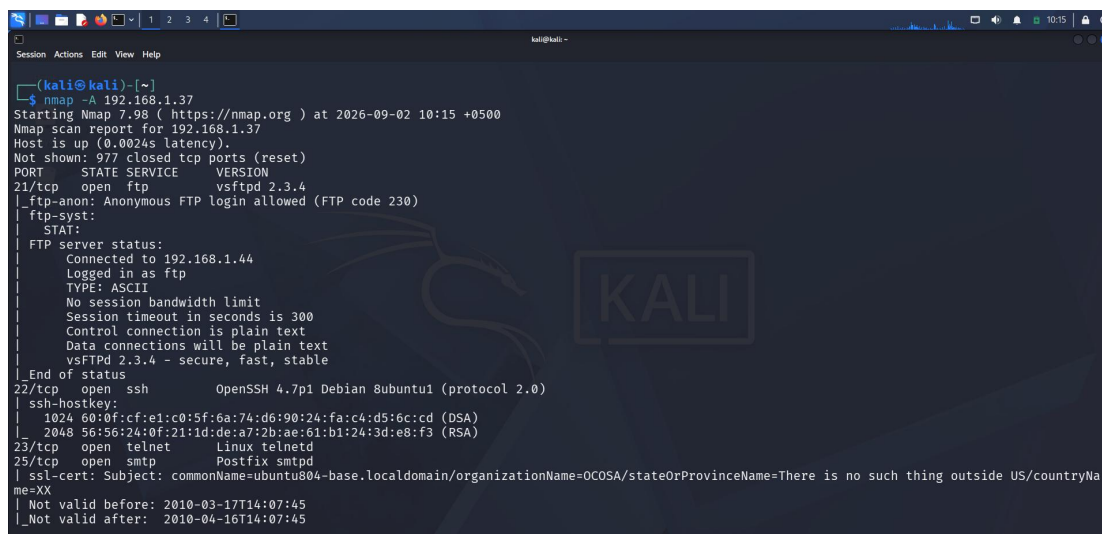
In the aboe Figure nmap **-sV 192.168.1.37** was executed. This command give us port number the services and there versions.

Open Ports and Services

S.No	Port Number	Service	Version	Status
1	21	ftp	vsftpd 2.3.4	open
2	22	ssh	OpenSSH 4.7p1 Debian 8ubuntu1	open
3	23	telnet	Linux telneted	open
4	25	smtp	Postfix smtpd	open
5	53	domain	ISC BIND 9.4.2	open
6	80	HTTP	Apache httpd 2.2.8 ((Ubuntu) DAV/2)	open
7	111	rpcbind	2 (RPC #100000)	open
8	139	netbios-ssn	Samba smbd 3.X -4.X (workgroup:WORKGROUP)	open
9	445	netbios-ssn	Samba smbd 3.X -4.X (workgroup:WORKGROUP)	open
10	512	exec	Netkit-rsh rexecd	open
11	513	login		open

12	514	tcpwrapped		open
13	1099	java-rmi	GNU classPathgrmiregistry	open
14	1524	bindshell	Metasploitable root shell	open
15	2049	nfs	2-4 (RPC #100003)	open
16	2121	ftp	PROFTPD 1.3.1	open
17	3306	mysql	MYSQL 5.0.51a-3ubuntu5	open
18	5432	postgresql	POSTGRESQL DB 8.3.0 - 8.3.7	open
19	5900	vnc	VNC (protocol 3.3)	open
20	6000	X11	(access denied)	open
21	6667	irc	UnrealIRCd	open
22	8009	ajp13	Apache Jserv (Protocol v1.3)	open
23	8180	http	Apache Tomcat/Coyote JSP engine 1.1	open

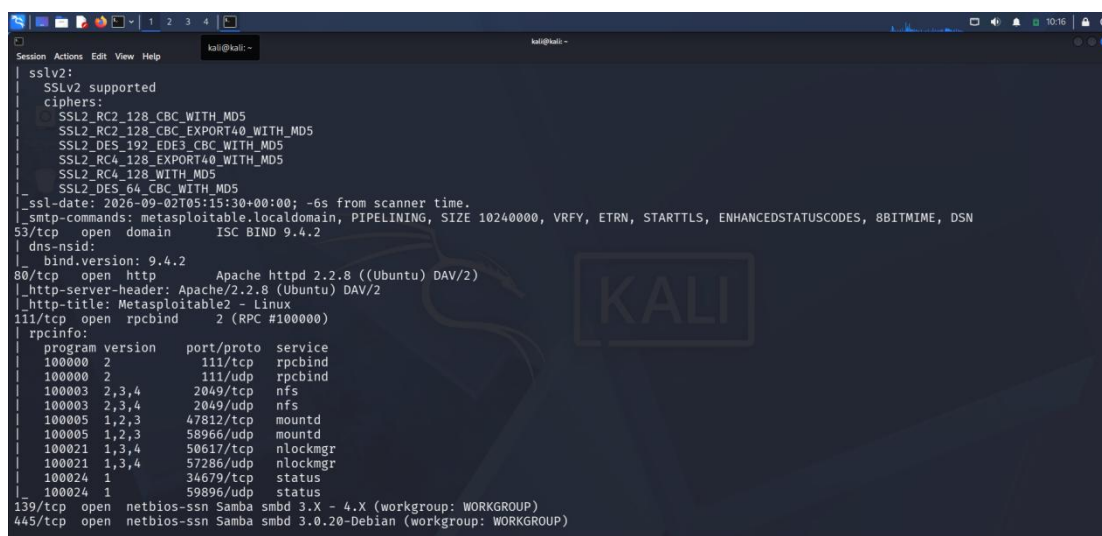
Aggressive Scan



```
(kali@kali)-[~]
$ nmap -A 192.168.1.37
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 10:15 +0500
Nmap scan report for 192.168.1.37
Host is up (0.0024s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.1.44
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
```

Figure 1.3

Nmap Aggressive Scan technique is used which will cover service version Os detection traacroute and nmap scripting engine scripts. In Figure 1.3 Port 21 23 25 are shown with ththeir service version.



```
sslv2:
|_SSLv2 supported
|_ciphers:
|_  SSL2_RC2_128_CBC_WITH_MD5
|_  SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_  SSL2_DES_192_EDE3_CBC_WITH_MD5
|_  SSL2_RC4_128_EXPORT40_WITH_MD5
|_  SSL2_RC4_128_WITH_MD5
|_  SSL2_DES_64_CBC_WITH_MD5
|_ssl-date: 2026-09-02T05:15:30+00:00; -6s from scanner time.
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_  program version port/proto service
|_  100000 2 111/tcp rpcbind
|_  100000 2 111/udp rpcbind
|_  100003 2,3,4 2049/tcp nfs
|_  100003 2,3,4 2049/udp nfs
|_  100005 1,2,3 47812/tcp mountd
|_  100005 1,2,3 58966/udp mountd
|_  100021 1,3,4 50617/tcp nlockmgr
|_  100021 1,3,4 57286/udp nlockmgr
|_  100024 1 34679/tcp status
|_  100024 1 59896/udp status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
```

Figure 1.4

In figure 1.4 the output is in continue of Aggressive Scan. In Fiure 1.4 the ports and their service version are outdated **Apache 2.2.8** web server (port 80), **ISC BIND 9.4.2** (port 53), **rpcbind/NFS** services (port 111), and an outdated **Samba 3.0.20** SMB service (ports 139/445) susceptible to remote code execution.

```
Session Actions Edit View Help
| 100024 1 34679/tcp status
| 100024 1 59896/udp status
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rshcd
513/tcp open login OpenBSD or Solaris rlogind
514/tcp open tcpwrapped
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
| mysql-info:
| Protocol: 10
| Version: 5.0.51a-3ubuntu5
| Thread ID: 9
| Capabilities flags: 43564
| Some Capabilities: Support41Auth, SupportsTransactions, ConnectWithDatabase, LongColumnFlag, SwitchToSSLAfterHandshake, SupportsCompression,
Speaks41ProtocolNew
| Status: Autocommit
| Salt: xsV4B/c%/YTW|)URx*d9
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
| Not valid after: 2010-04-16T14:07:45
| ssl-date: 2026-09-02T05:15:30+00:00; -6s from scanner time.
5900/tcp open vnc VNC (protocol 3.3)
| vnc-info:
| Protocol version: 3.3
| Security types:
| VNC Authentication (2)
```

Figure 1.5

In **figure 1.5** results are critical system entry points, including an unauthenticated root bindshell on port 1524 alongside legacy remote access services like exec and rlogin. It also exposes legacy database and desktop management instances, specifically outdated MySQL 5.0.51a, PostgreSQL 8.3, and an unencrypted VNC remote desktop server.

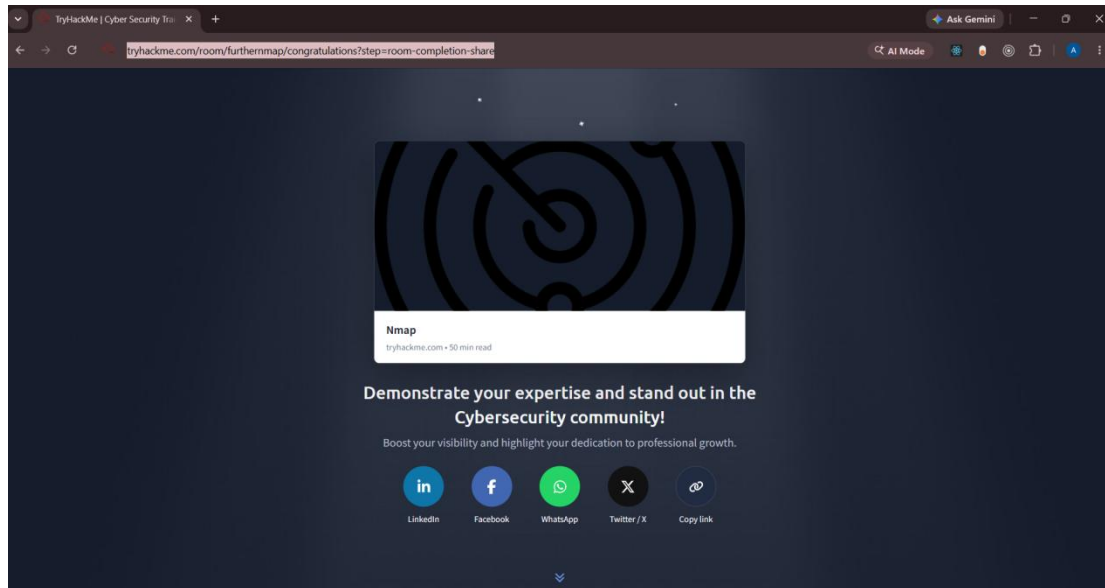
```
Session Terminal Emulator (Use the command line)
6000/tcp open x11 (access denied)
6667/tcp open irc UnrealIRCd
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
| ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
| http-server-header: Apache-Coyote/1.1
| http-title: Apache Tomcat/5.5
MAC Address: 08:00:27:9C:53:E5 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb2-time: Protocol negotiation failed (SMB2)
| smb-security-mode:
| account_used: <blank>
| authentication_level: user
| challenge_response: supported
| message_signing: disabled (dangerous, but default)
| clock-skew: mean: 59m54s, deviation: 2h00m00s, median: -6s
| nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb-os-discovery:
| OS: Unix (Samba 3.0.20-Debian)
| Computer name: metasploitable
| NetBIOS computer name:
| Domain name: localdomain
| FQDN: metasploitable.localdomain
| System time: 2026-09-02T01:15:22-04:00
```

Figure 1.6

In **Figure 1.6** it is shown that open web application services including **UnrealIRCd** (port 6667) and **Apache Tomcat/5.5** (port 8180). Host discovery scripts confirm the target is running an old **Linux 2.6.x** kernel on VirtualBox, with SMB security settings revealing **disabled message signing** and anonymous user access.

TryHackMe Nmap Room



TryHackMe nmap room was successfully completed. This room provides me hands-on experience with different network or target scanning techniques such as host discovery port scanning, enumeration, stealth scan, NSE scripts for vulnerability scanning, tcp scan, xmass scan, and so on.

Conclusion

In this task, I learned how to use Nmap for network scanning and reconnaissance. I performed host discovery, port scanning, service version detection, and aggressive scanning to identify open ports and running services on the target machine. I also completed the TryHackMe Nmap room, which improved my practical understanding of different Nmap scanning techniques. This activity enhanced my skills in using Nmap for ethical hacking and network security assessments.